

AI in Manufacturing and Industrial Operations

When the model is wrong, someone is under the machine

Industrial IoT

Predictive maintenance

Alert fatigue and drift

Safety authority

Finance priced the errors. Here the errors are physical.

Same five-stage method, same human-authority rule. What is new is time, degradation, and data that comes off equipment instead of out of a database.

I	Where AI runs in a plant, including IoT and data streams	18 min
II	Case 1: the maintenance decision	15 min
III	Lesson: anomaly detection in five stages	20 min
IV	Does the alert help a person? Alert fatigue and drift	12 min
V	Safety authority and honest ROI	12 min
VI	Wrap and the improvement-proposal brief	5 min

SECTION I · 18 min

A plant runs on equipment, and equipment becomes data

Five families of work, then one short path from sensors and controllers to a continuous stream the model can use.

Five families of work run a plant

FAMILY	THE QUESTION BEING ANSWERED	TONIGHT
Maintenance	Which machines need service before they fail, and which can safely wait?	Tonight's case
Quality inspection	Is this part good? Decided in under a second from a camera image.	Module 3 skill
Process control	What should temperature, pressure, and speed be right now?	—
Planning and scheduling	Which job runs on which line, in what order, with which people?	—
Safety and procedure	What does the official procedure require before anyone touches this machine?	—

Tonight is one case, maintenance. Quality inspection is a skill you already own from Module 3; the other three are real, large, and out of scope.

There are four ways to decide when to service a machine

STRATEGY	WHAT IT MEANS	WHAT IT COSTS YOU
Run to failure	Use the machine until it breaks, then fix it.	Unplanned downtime, at the worst possible moment.
Scheduled / preventive	Service every machine on a fixed calendar, healthy or not.	Servicing machines that were fine, and still missing the ones that fail early.
Condition-based	Watch one sensor and act when it crosses a fixed limit.	One limit on one sensor cannot tell a hot day from a hot machine.
Predictive (tonight)	Learn what normal looks like across several sensors together, and flag the machine that is drifting away from it.	Building it, tuning it, and every false alarm that sends a technician to a healthy machine.

Quality inspection: you already know how to build this

Same architecture as the chest X-ray

A camera photographs each part. An image model, the same kind you met on the chest X-ray in Module 3, calls it acceptable or defective, or draws a box around the flaw. It learns from labeled examples.

Same quality gate, same authority rule

The model does not scrap the part on its own. It routes borderline parts to a human inspector, and someone owns the decision to reject a batch and stop a line.

What is genuinely harder here

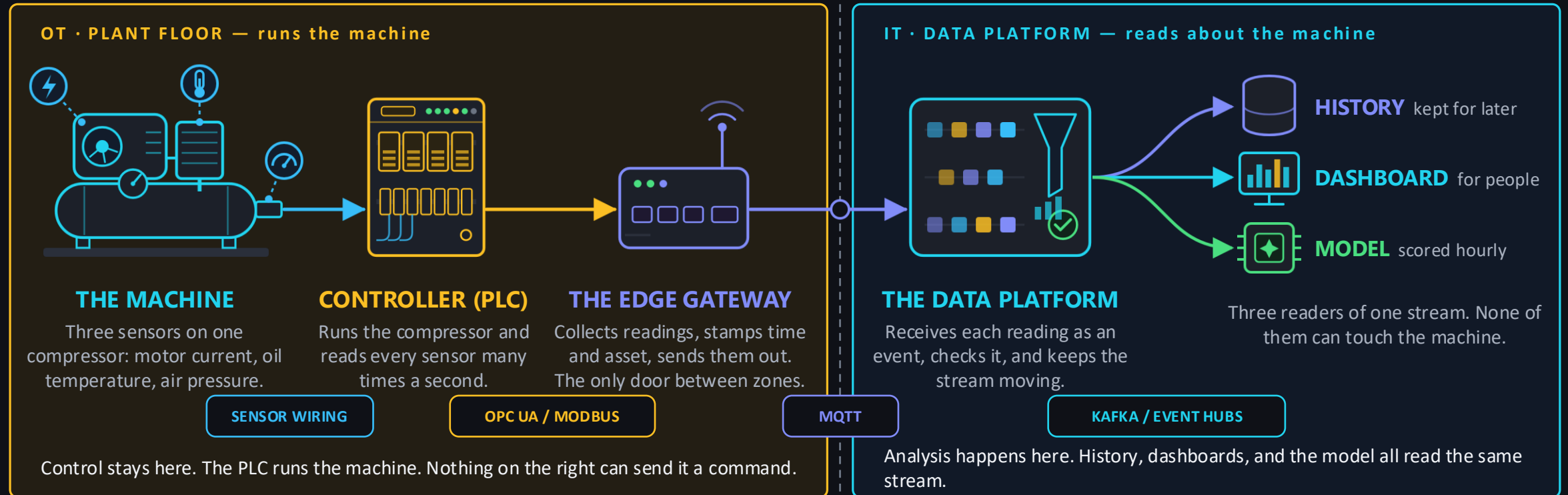
Defects are rare, so you have thousands of good parts and a handful of bad ones. Lighting, part position, and camera focus change the images more than the defects do. And a new product means a new dataset.

Why it is not tonight's case

It would repeat a skill rather than add one. Tonight's case brings something Module 3 could not: a machine that slowly degrades over time, with no label saying when.

Industrial IoT: a machine becomes a stream of readings

Readings travel from the plant floor to the data platform. Commands never travel back from the model to the machine.



Representative pattern with example technologies, following the common Purdue / ISA-95 separation of plant control from enterprise IT. Not a record of how the MetroPT train was wired.

A data stream is a sequence of timestamped events

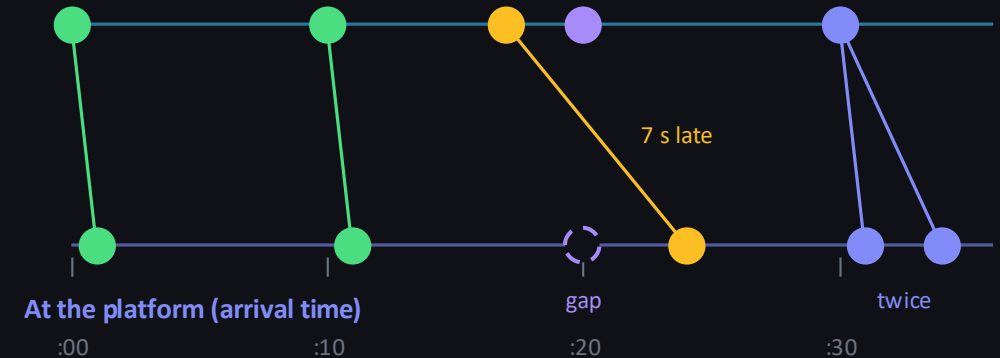
Instead of waiting for a finished table, the platform receives one small record at a time while the compressor runs.

ONE EVENT = ONE SMALL RECORD

EVENT TIME	ASSET	SIGNAL	VALUE	QUALITY
12:00:00	CP-01	Motor current	5.12 A	Good
12:00:10	CP-01	Oil temperature	58.4 °C	Good
12:00:17	CP-01	TP3 pressure	8.12 bar	Arrived 7 s late
12:00:20	CP-01	Motor current	—	Never arrived
12:00:30	CP-01	Oil temperature	58.4 °C	Arrived twice

SAME EVENTS ON TWO CLOCKS

At the machine (event time)



Late

Measured at :17, received at :24. It keeps its event time of :17.

Missing

A reading that never arrives leaves a gap. Do not silently fill it.

Duplicate

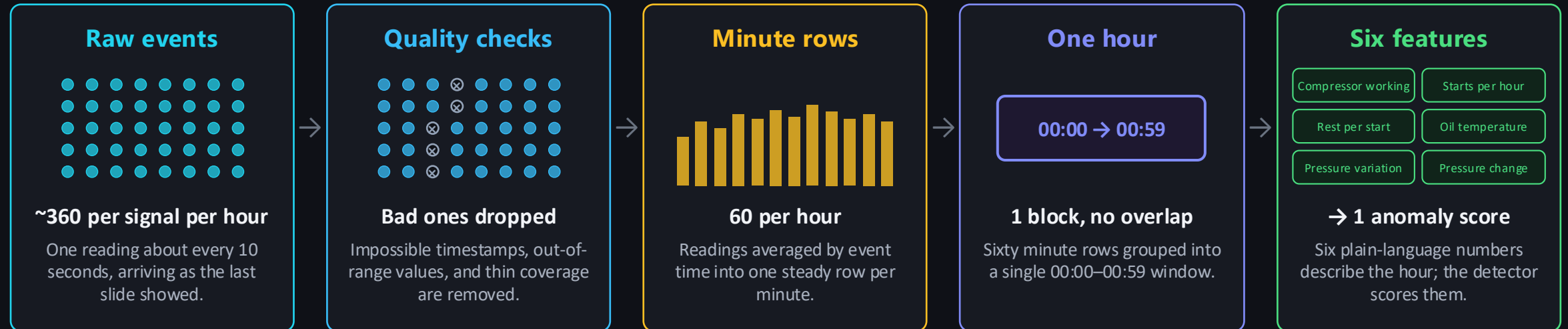
A retry can deliver one reading twice. Count it once, or averages drift.

Out of order

A later reading can arrive first. Sort by event time, never by arrival.

The model does not score raw sensor messages

One hour of readings is checked and summarized into six numbers before the detector scores it.



Why the score waits an hour



00:00 hour begins — nothing to score yet

01:00 hour complete → six features → score

The 00:00 hour holds readings from 00:00 to 00:59, so its score exists at 01:00. That is the hour the F4 lesson subtracts from apparent lead time.

What the classroom simulator does

It replays saved MetroPT readings hour by hour through this same sequence: readings arrive, six numbers, one score, one line. It is not connected to a live train and never sends anything back.

MetroPT-3 samples irregularly at roughly 10-second intervals; the lesson uses minute averages and builds one feature row per hour.

SECTION II · 15 min

Case one: a compressor on a train

Real sensor data from a Metro do Porto train compressor. Seven months, 1.5 million readings, four documented air leaks, and no column that says healthy or faulty.

The decision: which machine needs a closer look?

The problem

A maintenance team has more equipment than technician hours. Which compressor is behaving unusually enough to investigate?

What the model outputs

One anomaly score per hour. Higher means more unusual compared with earlier reference data — not a probability of failure.

What actually happens

At or above the cutoff, the system requests review. A planner and technician inspect the equipment and decide what to do.

A flag means: take a closer look

It does not mean the machine is broken. The model does not diagnose a fault, authorize work, or certify safety.

The notebook's detector (Isolation Forest) learns what a normal hour looks like. Nobody tells it which hours were failures; the maintenance reports are used only afterward, to check it.

Real equipment data, and the sensors say ordinary things

1.5 million readings from one compressor across seven months, February to September 2020. Every sensor has a plain physical meaning a technician would recognize.

WHAT THE SENSOR MEASURES

IN PLAIN LANGUAGE

Motor current

Is the compressor off, idling, or working hard right now?

Panel and reservoir pressure

How much compressed air is available, and how fast it is being used up.

Oil temperature

How hot the unit is running. A machine that never rests runs hotter.

Compressor on/off signal

When each working burst starts and stops — the machine's rhythm.

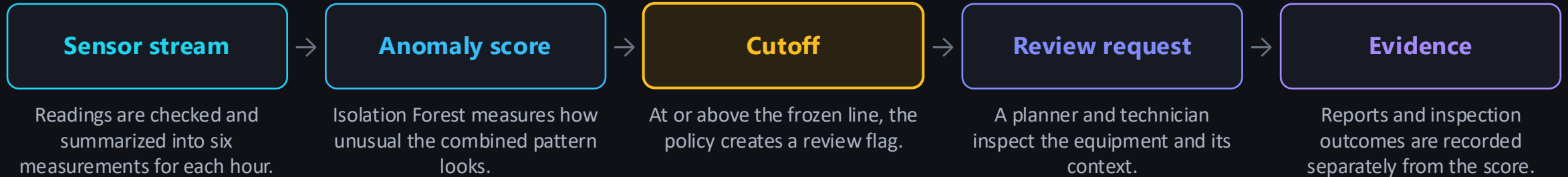
1.5M readings · 7 months

~10 s sampling, irregular

4 documented air leaks · no labels

MetroPT-3, Metro do Porto air production unit (UCI dataset 791, CC BY 4.0). Readings and failure dates verified first-party; the four failures come from the operator's own documentation, not from a label column.

The whole workflow, and only one box is the model



Reports help us check the model; they did not train it

They show when an event was documented. No matching report does not prove the machine was healthy.

What the model never does

It does not diagnose the fault, stop the machine, order a part, or close its own review flag.

What happens when scoring is unavailable

Missing or insufficient readings produce a visible hold. The established maintenance process remains in place.

An air leak makes the compressor work harder to hold pressure

Understand the physics first and the six numbers write themselves.

Air escapes

A leak develops somewhere in the compressed-air system.



Pressure falls faster

The tank empties sooner than it should after each burst.



It restarts sooner

It rests less between bursts and works for longer.



It runs hot

A machine that never rests heats up and stays hot.

The six numbers the notebook builds for every hour

Compressor working

share of the hour working hard

Starts per hour

how often it kicks in

Rest per start

minutes off between starts

Oil temperature

average for the hour

Pressure variation

how much tank pressure varies

Pressure change

how fast it moves while resting

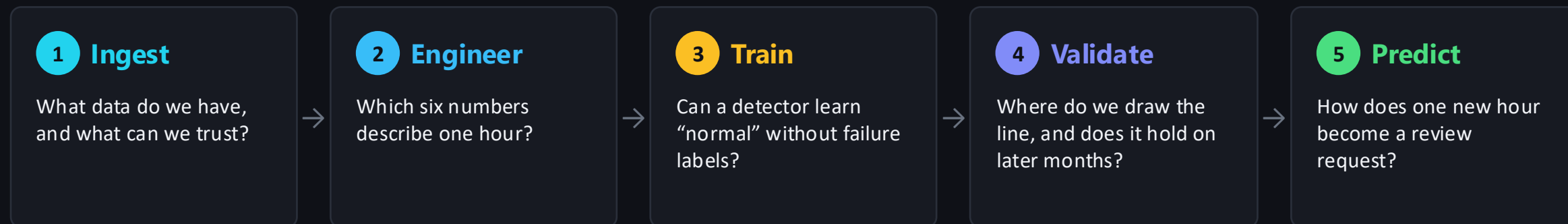
Colors match the step of the story each number measures. Same six names as the notebook.

Why plain-language numbers are worth the effort

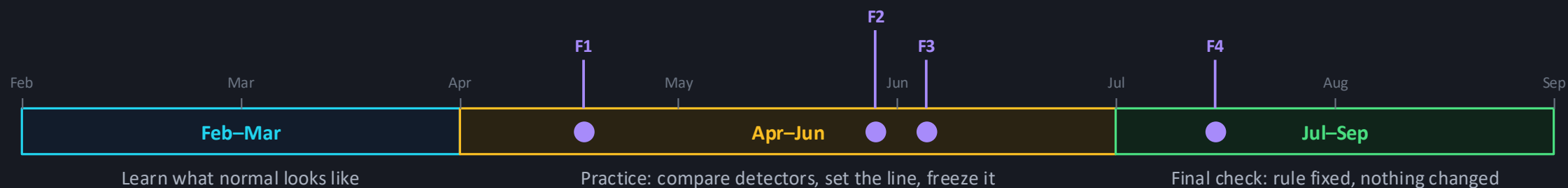
When an alert fires, the planner sees “this unit rested four minutes between bursts instead of eighteen, and ran nine degrees hot.” That is something a technician can argue with, check with a thermometer, and act on. “Sensor 7 was elevated” is none of those.

Open the lesson: anomaly detection in five stages

[01_pdm_build.html](#)



How the seven months are used



Three leaks fall in the practice months, one in the final check. The line is chosen on practice months and frozen before July, so the final check is honest.

After the lesson: what the final check shows

1 of 1

Reported leak caught

F4, the only leak in July–September, was flagged before it was reported.

27 of 1,224

Hours flagged

About 2% of the scored hours, the same share chosen in the practice months.

4

Extra callouts

Groups of flagged hours with no matching report. To investigate, not proven false.

13.5 h

Warning time

From the flagged hour's score at 01:00 to the reported start at 14:30.

How a score becomes a flag

The line is 0.679284. Below it, no flag. At or above it, ask a person to look. The number is a position on the line, not a 68% chance of failure.

What the reports can tell us

Whether a flag overlaps a documented leak, and whether the flag came first.

What the reports cannot tell us

That every unreported hour was healthy, or an accuracy percentage from a single leak.

SECTION IV · 12 min

Does the alert help a person?

A useful flag arrives early enough to act on, creates a manageable review queue, and stays meaningful when operating conditions change.

The cutoff decides how much review work exists

The model gives every hour a score. A person chooses where the line goes, and that choice sets how many hours land in front of a technician.



Line lower

More hours flagged. More possible leaks enter the queue, and more trips to machines that were fine.

Line as selected

Set so about 2% of practice hours were flagged, then frozen. Final check: 27 hours, 1 leak caught, 4 extra callouts.

Line higher

A cleaner queue. The 13.5-hour warning for F4 could disappear with it.

If operating conditions change, alert volume can change

Each new hour is compared with February and March. When alerts rise, there are three possible reasons.

The machine changed

A real fault is developing. This is the case the detector was built to catch.

The work changed

A busier timetable, a hotter season, a different duty. Normal moved; the machine did not.

A sensor changed

Recalibrated, drifting, or dropping readings. The data moved; nothing else did.

Find out which one before moving the line

More alerts do not prove more faults. Look at missing data, sensor readings, workload, and what technicians found, together. Moving the line first hides the cause.

Watch the inputs

Coverage, missing hours, and the usual range of the six numbers, not only the alert count.

Watch the workflow

Reviews, technician findings, missed leaks, and whether the queue is still usable.

F1–F4 are four documented leaks, not four model classes

All four leaks were flagged while they were happening. Only F4 was flagged before it was reported.

● model's first flag (score ready)

◆ leak in the maintenance report

WARNING

F1 Apr 18, 00:00
First flagged hour 06:00; its score exists at 07:00, seven hours after the report
No warning

F2 May 29, 23:30
First flagged hour 05:00 next morning; score exists at 06:00, 6.5 hours after
No warning

F3 Jun 5, 10:00
The flagged hour is the reported hour; its score exists at 11:00
No warning

F4 Jul 15, 14:30
Flagged hour 00:00; score exists at 01:00
About 13.5 hours

One useful warning does not prove the model can forecast every leak

F1–F3 sit in the months used to choose the model, so F4 is the only honest test, and a report says when a leak was written down, not that every other hour was healthy. More leaks, more compressors, and inspection outcomes are needed before claiming real performance.

SECTION V · 12 min

Who decides it is safe to work on the machine?

Tonight's system may not start, stop, or clear a machine. That boundary is not modesty about model quality. It is where the authority already sits, and it does not move because a score got better.

A score cannot supply machine context or authority

Safety-critical systems come in two kinds, and the difference is not how accurate they are. It is whether they are wired to the equipment.

Advisory — tonight's system

The output is information for a person: an anomaly flag with the six numbers behind it. The machine keeps doing what it was doing.

Interlocked — not tonight's system

The output is wired to the equipment: it trips, locks out, or refuses to start. Interlocks are real, and they are governed by functional-safety standards and certification — a different discipline from this course.

Two contributions to one safe workflow

MODEL

SCORE + FLAG

The model applies one measurement rule consistently. The responsible person checks evidence, applies the actual procedure, and owns the decision.

PERSON ▲ AHEAD

CONTEXT + DECISION

On the later test, the review policy costs more

THE THREE NUMBERS

HOW IT IS PRICED

WHY IT DECIDES THE ANSWER

A missed failure

Hours of running fault at a cost per hour, plus a large penalty once a unit is pulled out of service.

The biggest number, and the one you know least well.

An extra review

One technician review without matching report evidence, at a flat assumed cost.

Visible workload, but not proof that the equipment was healthy.

The baseline

Never alerting, and inspecting on a calendar — both priced with the same assumptions.

Doing nothing has a cost. So does the calendar. Neither is zero.

The selected Isolation Forest and cutoff, checked on July–September

NO-ALERT BASELINE ▲ AHEAD

\$5,400

FOREST REVIEW POLICY

\$6,800

This test contains one reported event and four extra grouped callouts against the reports. Every dollar is a classroom assumption; no figure comes from Metro do Porto.

Five ways these projects fail beyond the model

- 1 The incoming pattern changes and nobody investigates why**
Workload, schedules, sensors, or equipment condition can move the scores. Retuning first can hide the cause.
- 2 The review queue exceeds the team's capacity**
Flags accumulate faster than people can investigate them, so useful signals can be delayed or overlooked.
- 3 Inspection outcomes are not recorded**
Without findings linked to flags, the system cannot be evaluated, improved, or defended.
- 4 The data never gets off the plant floor**
Controllers and historians sit on isolated networks for good security reasons. Months disappear into plumbing, and the project is judged on a schedule that assumed modeling was the hard part.
- 5 The people who act on it were not part of the design**
If users cannot inspect the evidence or understand the next step, the output will not fit the real workflow.